MARK SCHEME · TEACHER COPY

12BR · Topic 14: Communication and Internet Technologies

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 35

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · Oct/Nov 2025 Paper 31, original Q3 (a),(b) [9 marks]

Question	Answer	Marks
1(a)	**One** mark per mark point (**Max 5**) **One** mark per mark point for **purpose** of Application Layer (**Max 3**) MP1 To provide services / interface with the user // access to applications, for example login, file transfer, network file access, email, etc MP2 To provide mechanisms for securing communication e.g. encryption/authentication MP3 To define/provide protocols used to allow the exchange of data/communication // to contain programs that exchange data MP4 Error detection and recovery mechanisms to handle application specific errors **One** mark per mark point for **purpose** of Transport Layer (**Max 3**) MP5 To provide logical communication between applications running on different hosts // To ensure that data is delivered to the correct application process on the destination machine MP6 To ensure error-free, end-to-end delivery of data between a source and a destination, in sequence // to provide error recovery techniques such as error detection codes and automatic repeat request MP7 To break data into segments when sent and to reconstruct when received // To reassemble segments at destination MP8 To regulate network connections // to provide flow control mechanisms to prevent data loss.	**5**
1(b)	**One** mark per mark point (**Max 4**) MP1 Data are broken into equal sized packets MP2 Data packets have headers containing information such as the IP addresses of the sender and receiver MP3 Each packet of data is sent independently to the destination // Packets don't necessarily follow the same route MP4 Each packet is sent via the most optimum path available MP5 Packets don't necessarily arrive in the order they were sent // Packets are reconstructed at the destination in the correct order MP6 Missing / damaged packets are re-sent	**4**

Question 2 · May/June 2026 Paper 32, original Q2 [4 marks]

Question	Answer	Marks	Guidance
2	Examiners must award **whole marks** (not half marks, or other fractions).		

2	**One** mark for each correct word in the correct place (**Max 4**) four packets link interconnects The TCP/IP protocol suite consists of **four** layers which can be thought of as a stack. It is designed to send **packets** of data across the internet. It is also used as a communications protocol to ensure accurate delivery of data over private networks. The lowest layer of the stack is the **link** layer. This is the layer that **interconnects** nodes in the network.	**4**	

Question 3 · Oct/Nov 2025 Paper 32, original Q3 (a),(b) [5 marks]

Question	Answer	Marks
3(a)	**Two** from • Application Layer • Transport Layer • Internet Layer • Link Layer	**1**

3(b)	**One** mark per mark point (**Max 4**) MP1 The TCP/IP suite can be viewed as layers within a stack MP2 Each layer can only accept input from the next higher or the next lower layer/adjacent layer MP3 The user/sender/computer interfaces with the top layer/application layer to send/receive a message MP4 The message when sent passes from the top layer/application layer to the bottom layer/link layer MP5 The message when received passes from bottom layer/link layer to the top layer/application layer MP6 The link layer interfaces directly with the network and sends/receives the message.	**4**

Question 4 · Oct/Nov 2025 Paper 32, original Q4 [4 marks]

Question	Answer	Marks

4	**One** mark for each benefit (**Max 2**) MP1 Suitable for long continuous transmission once the connection is made, it is available until the end of the transmission MP2 No time lost in resending/rearranging packets as no loss of packets or out of order packets MP3 Steady/high rate of transmission because the whole of the bandwidth is available MP4 Data loss unlikely as all data follows the same path **One** mark for each drawback (**Max 2**) MP5 Delays due to dedicated connection required to be set up/established before transmission can begin MP6 The dedicated connection cannot be used to transmit any other data MP7 System resources may be underutilised/inefficient/not very flexible because the bandwidth can't be shared/high bandwidth is required // Bandwidth may be wasted // May send empty frames // the circuit is always there whether or not used MP8 There are no alternative routes if there is a failure or fault on the line MP9 Reduced security due to use of single path MP10 Scalability is difficult	**4**

Question 5 · May/June 2026 Paper 31, original Q4 [4 marks]

Question	Answer	Marks	Guidance
5 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		

Total: 35 marks

5	**One** mark for each correct line connecting a TCP/IP layer to its description **(Max 4)**. **TCP/IP layer** — **description** Application Transport Internet Link ensures packets arrive in sequence without errors provides remote access to files on a host server to access and manage files stored in a remote location interconnects nodes or hosts in a network, encapsulates IP packets into frames and ensures correct protocols are followed identifies the intended network and host. contains all the programs that exchange data and includes protocols: HTTP, FTP and POP3	**4**	

Question 6 · Oct/Nov 2025 Paper 33, original Q4 (a),(b) [9 marks]

Question	Answer	Marks

6(a)	**One** mark per mark point (**Max 5**) **One** mark per mark point for **purpose** of Internet Layer (**Max 3**) MP1 To identify the intended network and host. MP2 To add header containing IP addresses. // To address packets with their source and destination IP Addresses. MP3 To prepare packets for delivery by formatting them into datagrams. MP4 To route datagrams through the optimum route over a network. **One** mark per mark point for **purpose** of Link Layer (**Max 3**) MP5 To prepare the next hop by managing the link between two directly connected devices. // To identify and move traffic across local segments. MP6 To format datagrams into frames for transmission. MP7 To identify network protocols in the packet header. // To ensure correct network protocols are/is followed. MP8 To deliver frames to the receiving network. // To receive frames from the sending network. // To map IP addresses to MAC physical addresses. MP9 To provide error checking/error correction/recovery/reconstruction MP10 including resend requests.	**5**
6(b)	**One** mark for each correct marking point **(Max 4**) MP1 The router reads the IP address of the destination from the packet header. MP2 A router uses a routing table to find information … MP3 … about e.g., available hops / netmask / gateway used / adjacent routers / the status of the routers along the route. MP4 The router determines the next hop / optimum route. // The router sends the packet on its next hop. MP5 The router manages the hop counter // The hop counter is reduced by 1 every time the packet passes a router.	**4**

6											